Improved A Algorithm for Autonomous Vehicle Path Planning under Post-disaster Rescue Scene*

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***Abstract*—Mobile robots play a crucial role in various applications, such as explosion prevention, warning systems, and post-disaster rescue missions, owing to their small size and flexible movement capabilities. Traditional mobile robots often employ a single motion mode, which can be insufficient in adapting to the complex and dynamic post-disaster rescue environments. Composite mobile robots, equipped with multiple motion capabilities, have gained popularity in such scenarios. This paper focuses on wheel-legged robots as carriers and proposes a heuristic function that combines distance and energy consumption constraints to improve the A* algorithm. The goal is to achieve effective global path planning in the challenging post-disaster rescue conditions. The environmental features are represented using distance grid maps and energy grid maps. Experimental results demonstrate that the improved algorithm reduces path planning time and successfully balances optimal path selection with energy-efficient planning in typical post-disaster rescue conditions.**

Keywords—Mobile robots, A* algorithm, Heuristic approaches, Multiple motion capabilities

I. INTRODUCTION

After natural disasters strike, the underground environment often undergoes significant changes in its geometric structure[1], leading to uncertainty and variability in the disaster information. To address these critical issues, researchers have turned to mobile robots as a promising solution in the field of emergency rescue. Mobile robots offer distinct advantages, including their compact size and flexible mobility, making them invaluable tools for post-disaster rescue operations.

Traditional mobile robots face significant limitations when navigating through rugged terrain and complex narrow underground spaces[2]. Their inability to efficiently penetrate underground shelters with these challenges hinders their ability to perform unknown environment exploration tasks effectively. To address these limitations, our paper introduces a novel solution: the wheel-track-leg autonomous unmanned vehicle[3]. This innovative vehicle offers a versatile approach to handling complex working conditions by seamlessly transitioning between three fundamental motion modes: wheeled, tracked, and legged. These modes enable the vehicle to adapt to the terrain, selecting the most suitable motion mode for the task at hand, thereby facilitating obstacle-crossing and exploration tasks in challenging environments.

This paper is dedicated to addressing these challenges by focusing on the global path planning problem. Our goal is to identify and optimize an optimal path from the initial position to the desired goal position within environments that are either entirely unknown or partially known. By doing so, we aim to provide a practical solution for navigation in scenarios where local path planning alone falls short.

Global path planning[4] algorithms encompass various methods for finding optimal paths in unknown or partially known environments. Dijkstra's algorithm[5], a classical approach, employs a greedy strategy to iteratively identify the nearest unexplored cells from the starting point to the endpoint. However, it faces the challenge of blind global search[6], which results in substantial computational resource and time consumption. To address the limitations of Dijkstra's algorithm, Michael[7] introduced the Probabilistic Roadmap Method (PRM), which discretizes continuous space and leverages search algorithms like A* to find paths on the roadmap. Steven[8] proposed the Rapidly-exploring Random Tree (RRT) approach, which generates a random tree by sampling additional leaf nodes from certain points. While PRM and RRT offer probabilistic completeness, they do not guarantee the optimal path and are computationally demanding due to their random search nature. Peter[9] introduced the A* algorithm, which incorporates a heuristic function by adding the distance to the endpoint. Although A* enhances search efficiency, it still suffers from the problem of replanning. In the realm of UAV[10] 3D path planning, Zhang[11] improved efficiency by introducing a weighting factor to balance different costs in path planning and by employing the virtual terrain method, which considers the UAV's performance to constrain the search space in the temporal dimension.

The A* algorithm is a widely recognized graph search path planning algorithm known for its speed, performance, and

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adaptability[12]. Its efficiency can be further improved by incorporating heuristic functions tailored to specific constraints. Given its versatility in handling complex scenarios, this paper selects the A* algorithm[13] as the foundational framework to develop a novel global path planning algorithm[14].

The primary contributions of this paper can be summarized as follows:

1) We enhance the A* algorithm by introducing a novel heuristic function that considers both distance constraints and energy consumption constraints. This tailored heuristic function is designed to facilitate efficient path planning for post-disaster rescue scenarios.

2) Leveraging the kinematic characteristics of the multi-motion modes available to the wheel-track-leg robot, we integrate obstacle information from the map environment to enable effective obstacle avoidance and obstacle-crossing actions. This is achieved by seamlessly switching between different motion modes, ensuring the robot can reach its target position while navigating complex terrains and obstacles.

3) Our experimental results confirm the effectiveness of the improved algorithm in reducing path planning time in typical post-disaster rescue conditions.

II. Robot and its Overall Architecture

The experimental platform used in this paper, as illustrated in Fig. 1, is a versatile wheel-track-leg robot prototype with multiple motion modes, including wheeled, tracked, and legged. The legged mode is primarily achieved through a swinging arm motion system, structurally designed with a triangular track and a single degree of freedom.

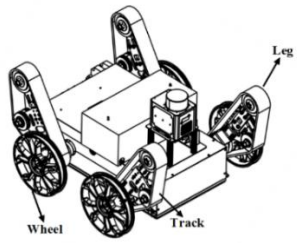


Fig. 1 Wheel-track-leg robot model

The robot is equipped with a 32-line LiDAR sensor located on its front side. In this study, point cloud information collected by the LiDAR serves as known environmental data. When the robot acquires this point cloud information, it is transmitted to the onboard CPU for point cloud alignment and downsampling.

A terrain analysis module processes the mapped data, generating distinct passable areas corresponding to wheeled, tracked, and legged motion modes. Global path planning is then conducted based on these segmented passable areas. The system architecture diagram of the robot is shown in the Fig. 2. As the robot advances and reaches the boundaries of different terrains, the mode switching module triggers the transition between motion modes, allowing the robot to seamlessly adapt to the terrain. The final control commands are transmitted to the robot's motors through CAN communication, enabling precise and coordinated movement in response to the chosen mode.

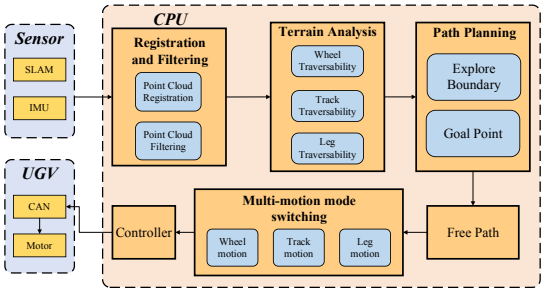


Fig. 2 System architecture diagram of the robot

III. Global Path Planning Algorithm

A. Grid Map And Adjacency Matrix

The robot's mission objectives vary depending on its battery status: When the robot's onboard battery is sufficiently charged, the primary objective is to complete the rescue mission in the shortest possible time. This prioritizes planning for the shortest path to the rescue location to ensure a swift response. In cases of low battery levels, the robot's primary consideration is to minimize energy consumption. This may involve planning for a return to the starting point or the rescue center with the aim of conserving energy resources while accomplishing the mission efficiently.

In this paper, we opt for grid maps as a convenient means of approximating the static environment. Grid maps offer ease of creation and maintenance, with the grid format capable of storing terrain parameter data within the local area. Within the framework of grid map-based path planning, we utilize the adjacency matrix as a fundamental component for robot path search.

The adjacency matrix serves as a numerical representation of the connections between positional nodes in the environment. By incorporating information related to distance and energy consumption into the adjacency matrix, we leverage graph search algorithms to achieve optimal path planning under different operational conditions.

In grid maps, the elements in the distance matrix $A_D(i,j)$ denote the Euclidean distance between that neighboring node i and j, i.e., $A_D(i,j)=s_{i,j}$, where $s_{i,j}$ denotes the Euclidean distance between the neighboring nodes i and j.

The distance grid map, as illustrated in Fig. 3, serves as a representation of the distance relationships among the robot's position nodes. In this map, various colors are used to signify different characteristics: Areas depicted in black indicate occupied regions, marking areas that are impassable for the robot. White areas in the grid represent wheeled passable regions, signifying spaces where the robot can utilize its wheeled motion mode. Blue areas denote tracked passable regions, indicating locations suitable for the tracked motion mode.

The grids in the map are labeled starting from the bottom-right corner, as depicted in the Fig. 3. Each grid is associated with a unique number that corresponds to its position in the distance adjacency matrix.

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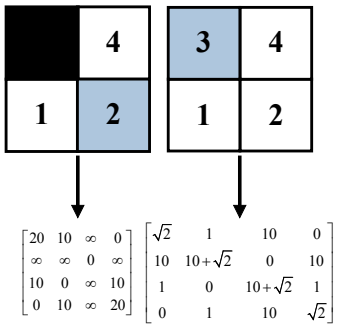


Fig. 3 Grid maps based on distance

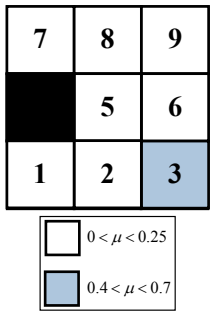


Fig. 4 Grid maps based on energy consumption

When neighboring areas are both wheeled or tracked passable, the robot can proceed without mode switching, eliminating the need to consider changes in trajectory length and increased energy consumption associated with mode switching.

For instance, consider the distance between grid 0 and grid 1. In cases where the neighboring grids represent a combination of wheeled and track-passable areas, the path selected may require mode switching for the robot to reach the target position. In our approach, mode switching is a primary process involving the robot's deceleration, cessation of movement, pendulum arm swinging, and transition to track or leg mode. Following this, the robot reaccelerates and resumes movement. During this primary process, the time required for mode switching is evaluated, resulting in an average value $\hat{t}$.

This estimated time, in combination with the set operating speed v_{set} , allows us to calculate the estimated cost of mode switching for the wheel-track-leg robot. The estimated cost of mode switching is defined as follows, where the set S denotes the same type of passable area:

$$A_D(i,j)=\begin{cases} s_{i,j}, i \wedge j \in S \\ s_{i,j}+v_{set}\hat{t}, i \vee j \in \bar{S} \end{cases} \quad (1)$$

In the grid map, the elements in the energy matrix $\mathrm{A}_E(i,j)$ are represented as the energy consumed by the robot to move from node i to node j :

$$\mathrm{A}_E(i,j)=0.5\mu_{i,j}s_{i,j} \quad (2)$$

$s_{i,j}$ denotes the Euclidean distance between neighboring nodes i and j , $\mu_{i,j}$ denotes the regional friction coefficient of neighboring nodes i, j and satisfies:

$$\mu_{i,j}=\mu_i+\mu_j \quad (3)$$

In the energy-based grid map shown in Fig. 4, various friction coefficients are employed to characterize the ruggedness of the grid environment, providing essential data for energy calculations in our path planning. In this representation, white grids indicate wheeled passable areas, with a low rolling friction coefficient typically below 0.25, reflecting their relative ease of traversal. The blue grids, on the other hand, represent tracked passable areas. The friction coefficient in these regions depends on factors such as track width, pattern, and the number of knots, typically falling within the range of 0.4 to 0.7.

These friction coefficients play a crucial role in determining energy constraints, ultimately influencing the robot's path planning decisions, ensuring efficient traversal in diverse terrain conditions.

B. Heuristic function construction

In the context of the graph search algorithm, the A* algorithm is a valuable tool for finding optimal paths and improving search efficiency. This is achieved by introducing a heuristic term that considers the distance information between the current position and the target position.

During each iteration of the algorithm, the cost function for the current extension point is determined. This is accomplished by calculating the cost of the distance from the starting point and an estimated surrogate value of the distance from the target point. The resulting evaluation function, denoted as:

$$f(n)=g(n)+h(n) \quad (4)$$

The evaluation function combines these factors to effectively guide the path planning process where $g(n)$ denotes the fixed cost from the current point to the starting point, and $h(n)$ denotes the estimated cost from the current point to the target point. In order to control the search speed and accuracy of the A* algorithm in larger scale maps, this paper introduces dynamic weight $w(n)$ to weight the estimated value of $h(n)$. The processing formula can be expressed as:

$$f(n)=g(n)+w(n)\cdot h(n) \quad (5)$$

In our algorithm, we utilize weight coefficients denoted by $w(n)$. The adjustment of these coefficients serves a dual purpose: For large-size maps, we increase the weight coefficient $w(n)$ to expedite the convergence of path planning results toward the target. This optimization is tailored to ensure quick responses in extensive environments.

Conversely, in small-size maps, we reduce the weight coefficient $w(n)$ to prioritize the accuracy of the search path. This adjustment guarantees that the search path remains precise in compact settings.

Thus the flow of the improved algorithm can be denoted as TABLE I:

TABLE I IMPROVED A* ALGORITHM

Algorithm 1: Improved A* path planning

Input: Map information ***M***, starting point $\boldsymbol{P_b}$, ending point $\boldsymbol{P_g}$, threshold for modifying weights ***H***.

Output: Collision-free paths connecting starting and ending points $\boldsymbol{P_f}$.

1. Adding the starting point $\boldsymbol{P_b}$ to the *OPEN* list.
2. Iterate through all the neighboring nodes of the current node and compare the size of the heuristic terms ***h(n)*** and ***H***.
3. $\boldsymbol{w(n)=w_h}$ when $\boldsymbol{h(n)>H}$, $\boldsymbol{w(n)=w_l}$ when $\boldsymbol{h(n)<H}$, $\boldsymbol{w_h}$ and $\boldsymbol{w_l}$ are the maximum and minimum values of ***w(n)*** respectively.
4. Calculate the minimum evaluation function node $\boldsymbol{P_m}$, remove it from the *OPEN* list and add it to the *CLOSE* list.
5. Determine if the next neighboring node $\boldsymbol{P_n}$ exists and if the *OPEN* list is empty.
6. The reverse-order outputs obtain the optimal paths $\boldsymbol{P_o}$ under distance constraints and energy consumption constraints.

From this, the heuristic function of the robot under the distance constraint can be expressed as:

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$$D(k-1,k)=\begin{cases}0.5Ms_{k-1,k}, k-1\wedge k\in S\\ 0.5M\eta_{k-1,k}s_{k-1,k}, k-1\vee k\notin \bar{S}\end{cases} \quad (6)$$

When the robot travels in the same passable area, there is no need to consider the distance change generated by mode switching, $D(k-1,k)$ denotes the cost of distance between neighboring nodes and M denotes a constant related to the robot mass, $\eta_{k-1,k}$ denotes the distance change coefficient between two neighboring nodes in different areas, and $s_{k-1,k}$ is the distance between two neighboring nodes.

Therefore, the heuristic function of the wheel-tracked legged robot from the current node to the goal node under the distance constraint can be expressed as:

$$F_D(k)=D\left(k,P_g\right) \quad (7)$$

The primary factors influencing the robot's path planning under energy constraints encompass the friction coefficient of the ground and the energy loss during mode switching. Among these factors, the ground's friction coefficient is a dominant consideration within the same motion mode of the robot.

We account for the physical characteristics of the work performed by the robot to counteract friction force and consider the impact of mode switching. As a result, we introduce a quantitative index to assess robot motion under energy consumption constraints that denoted as:

$$E(k-1,k)=\begin{cases}0.5M\mu_{k-1,k}s_{k-1,k}, k-1\wedge k\in S\\ 0.5M\mu_{k-1,k}s_{k-1,k}+\hat{\mathrm{E}}, k-1\vee k\notin \bar{S}\end{cases} \quad (8)$$

In the above equation, $E(k-1,k)$ denotes the value of energy lost between neighboring nodes $\mu_{k-1,k}$ is the friction coefficient between the two neighboring nodes, $s_{k-1,k}$ is the distance between the two neighboring nodes, and $\hat{E}$ is an estimated average energy consumption due to mode switching.

The energy cost of the robot from the starting point to the current node can be obtained based on the average energy consumption:

$$G_E(k)=G_E(k-1)+E(k-1,k) \quad (9)$$

In summary, the heuristic function of the wheel-tracked legged robot from the current node to the goal node under the energy consumption constraint can be expressed as:

$$F_E(k)=G_E(k)+E\left(k,P_g\right) \quad (10)$$

Through the synergistic combination of the heuristic function and the adjacency matrix, we achieve a robust quantitative representation of both the distance and energy consumption experienced by the robot throughout its path.

This combination integrates the heuristic function's capability to guide path planning with the adjacency matrix's capacity to account for distance and energy-related factors. The result is a quantitative representation that empowers energy-aware path planning, ensuring efficient and effective navigation while considering the robot's constraints.

IV. Experiments

The experimental scenario involves the placement of three fixed obstacles, as depicted in Fig. 5 (a) and Fig. 5 (b). Obstacle 1 and Obstacle 2 are cambered obstacles, while Obstacle 3 represents rugged terrain. These obstacles have a height of approximately 0.15 meters, and the robot is required to switch from wheel mode to track mode or leg mode to navigate through them.

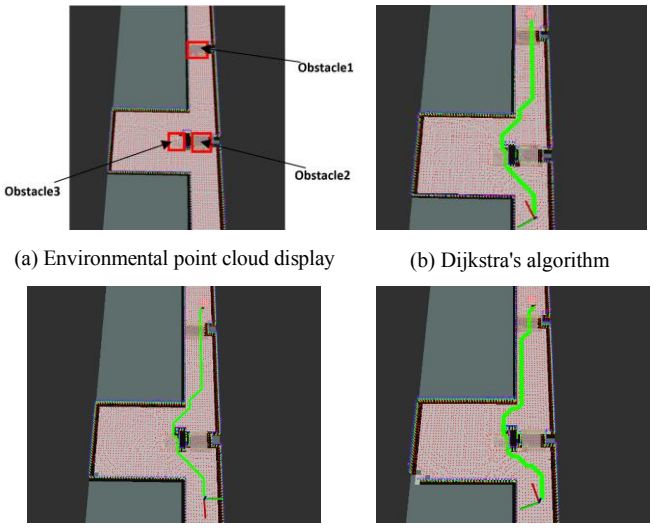
The Fig. 6 (a) displays the grid map generated by converting the collected LiDAR point cloud information. In this representation, the light red area signifies the explored passable regions, the black area represents impassable areas, and the grey area indicates unexplored territory.



(a) Obstacle 1 (b) Obstacle 2 and Obstacle 3

Fig. 5 Obstacles in the corridor

Additionally, the areas containing Obstacles 1, 2, and 3 are associated with a higher cost of passage compared to standard passable areas.



(a) Environmental point cloud display (b) Dijkstra's algorithm

(c) A* algorithm (d) The algorithms in this paper

Fig. 6 Obstacles and Path planning algorithm results

During the experiment, we recorded several key metrics, including the algorithm planning time, the robot's running time, the length of the robot's trajectory, and the energy loss.

The path planning results for the robot under the distance constraint are illustrated in Fig. 6 (b), (c) and (d). The green curve represents the global planning path.

TABLE II Indicators for distance constraints

Algorithm name	Serial number / average value	Planning time (ms)	Trajectory length (m)	Energy consumption (mAh)
Dijkstra algorithms	1	60.44	14.81	1390
	2	76.42	14.39	1255

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	3	63.85	15.07	1340
	average	**66.90**	**14.76**	**1328.33**
A* algorithms	1	55.01	14.11	1293
	2	42.83	14.72	1351
	3	45.41	14.27	1305
	average	**47.75**	**14.37**	**1316.33**
The algorithms in this paper	1	30.56	13.33	1458
	2	34.08	13.74	1365
	3	35.42	13.42	1424
	average	**33.35**	**13.50**	**1415.67**

Considering the cost of mode switching for the robot, the desired planning path when facing Obstacle 2 is to take a detour around the obstacle, opting for a slightly longer path with lower overall cost. In contrast, when the robot encounters Obstacle 1, with all passable paths in legged mode areas, the desired path is to switch the motion mode and traverse the obstacle. Obstacle 3 simulates a rugged ground area. This area can be crossed by switching to wheel mode or track mode. To mitigate the influence of randomness, we conducted three separate repetitions of each experiment group. The outcomes and evaluation metrics are summarized in the TABLE II.

Upon analyzing the experimental results, we observed that the improved algorithm significantly reduced planning time. Compared to Dijkstra's algorithm, it achieved a remarkable reduction of 50.01%, and in comparison to A*'s algorithm, it reduced planning time by 30.16%. Furthermore, concerning trajectory length, the improved algorithm demonstrated an 8.54% reduction in trajectory length compared to Dijkstra's algorithm and a 6.05% reduction compared to A*'s algorithm.

Similarly, we repeated the experiments 3 times for each experimental group under energy constraints to ensure reliable results. The results of the experiment are shown in the TABLE III: In terms of planning time, the improved algorithm outperformed Dijkstra's algorithm with a reduction of 38.78%, and it also exhibited a 14.62% reduction compared to the A* algorithm. Regarding the trajectory length, the improved algorithm showed a slight increase of 2.90% when compared to Dijkstra's algorithm, and a 1.65% increase compared to the A* algorithm. In energy consumption, the improved algorithm recorded a 6.27% reduction compared to Dijkstra's algorithm, but it exhibited a 6.67% increase compared to the A* algorithm.

TABLE III INDICATORS FOR ENERGY CONSUMPTION CONSTRAINTS

Algorithm name	Serial number / average value	Planning time (ms)	Trajectory length (m)	Energy consumption (mAh)
Dijkstra algorithms	1	62.41	16.33	1423
	2	70.72	16.28	1385
	3	64.39	16.02	1402
	average	**65.84**	**16.21**	**1403.33**
A* algorithms	1	58.05	16.77	1401
	2	43.76	16.15	1438
	3	39.82	16.31	1389
	average	**47.21**	**16.41**	**1409.33**
The algorithms in this paper	1	32.81	16.79	1316
	2	39.97	16.36	1333
	3	48.16	16.90	1297
	average	**40.31**	**16.68**	**1315.33**

V. CONCLUSION

In this study, we have delved into the application of the improved A* algorithm in the realm of path planning for autonomous unmanned vehicles, specifically addressing various environmental constraints such as distance and energy consumption.

The significance of this research lies in its provision of a comprehensive and pragmatic solution to real-world path planning predicaments. Our approach takes into account the intricate trade-offs among distance, energy consumption, and planning time, allowing for a balanced consideration of these critical factors. Future research endeavors may extending its application to even more intricate environments and scenarios.

REFERENCES

[1] M. Roberts, T. Apker, B. Johnson *et al*. Coordinating Robot Teams for Disaster Relief[C]//*FLAIRS Conference*. 2015: 366-371.

[2] M. Schneier and R. Bostelman, "Literature review of mobile robots for manufacturing," 2015.

[3] J. Gregory *et al.*, "Application of multi-robot systems to disaster-relief scenarios with limited communication," in *Field and Service Robotics: Results of the 10th International Conference*, 2016: Springer, pp. 639-653.

[4] P. Marin-Plaza, A. Hussein, D. Martin, and A. d. l. Escalera, "Global and local path planning study in a ROS-based research platform for autonomous vehicles," *Journal of Advanced Transportation,* vol. 2018, pp. 1-10, 2018.

[5] H. Wang, Y. Yu, and Q. Yuan, "Application of Dijkstra algorithm in robot path-planning," in *2011 second international conference on mechanic automation and control engineering*, 2011: IEEE, pp. 1067-1069.

[6] H. S. Lope and L. S. Coelho, "Particle swarn optimization with fast local search for the blind traveling salesman problem," in *Fifth International Conference on Hybrid Intelligent Systems (HIS'05)*, 2005: IEEE, pp. 245-250.

[7] S. M. LaValle, M. S. Branicky, and S. R. Lindemann, "On the relationship between classical grid search and probabilistic roadmaps," *The International Journal of Robotics Research,* vol. 23, no. 7-8, pp. 673-692, 2004.

[8] S. Karaman, M. R. Walter, A. Perez, E. Frazzoli, and S. Teller, "Anytime motion planning using the RRT," in *2011 IEEE international conference on robotics and automation*, 2011: IEEE, pp. 1478-1483.

[9] A. Gasparetto, P. Boscariol, A. Lanzutti, and R. Vidoni, "Path planning and trajectory planning algorithms: A general overview," *Motion and Operation Planning of Robotic Systems: Background and Practical Approaches,* pp. 3-27, 2015.

[10] M. Mueller, N. Smith, and B. Ghanem, "A benchmark and simulator for uav tracking," in *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*, 2016: Springer, pp. 445-461.

[11] Z. Qi, Z. Shao, Y. S. Ping, L. M. Hiot, and Y. K. Leong, "An improved heuristic algorithm for UAV path planning in 3D environment," in *2010 Second International Conference on Intelligent Human-Machine Systems and Cybernetics*, 2010, vol. 2: IEEE, pp. 258-261.

[12] L. Yang, J. Qi, D. Song, J. Xiao, J. Han, and Y. Xia, "Survey of robot 3D path planning algorithms," *Journal of Control Science and Engineering,* vol. 2016, 2016.

[13] C. W. Warren, "Fast path planning using modified A* method," in *[1993] Proceedings IEEE International Conference on Robotics and Automation*, 1993: IEEE, pp. 662-667.

[14] P. G. Luan and N. T. Thinh, "Hybrid genetic algorithm based smooth global-path planning for a mobile robot," *Mechanics Based Design of Structures and Machines,* vol. 51, no. 3, pp. 1758-1774, 2023.

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